CELLAR POS

COMPREHENSIVE SOFTWARE ARCHITECTURE, TECHNICAL SPECIFICATION & OPERATIONS MANUAL

Version 1.0.0 Production • Commercial Retail Solution • September 2026

1. EXECUTIVE SUMMARY & SYSTEM OVERVIEW

1.1 Business Context & Problem Statement: The retail wines and spirits sector in Kenya operates in a dynamic, high-volume trading environment. Retailers face frequent cellular network outages, extreme peak congestion on weekends and holidays, inventory shrinkage across multiple bottle sizes (250ml, 350ml, 750ml, 1 Litre), and daily reconciliation challenges reconciling physical cash drawers against Safaricom M-Pesa merchant till statements.

1.2 Core System Objectives:

- **100% Offline Autonomy:** Fully operational retail till requiring zero internet connection for catalog lookups, checkout ringing, digital receipting, or daily reconciliations.
- **High-Throughput Ringing:** Rapid barcode scanning and touchscreen-friendly interface designed for sub-5 second sales throughput.
- **Multi-Channel Receipts:** Digital receipts supporting 58mm/80mm ESC/POS thermal printing, direct WhatsApp customer dispatch, SMS formatting, and ASCII plaintext copy.
- **Shrinkage Control & Auditing:** Automatic stock deductions on sale, automatic stock restoration on voided sales, and mandatory-reason shrinkage logs.
- **Daily Dual-Tender Reconciliation:** Separate tracking and reconciliation for physical Cash Drawer versus M-Pesa Till Statement.

[KRA eTIMS VERSION 1.0 SCOPE NOTICE]
In accordance with Section 7 of the software requirements, Cellar POS Version 1.0 operates as an internal retail management till. Receipts generated are internal commercial records and are not transmitted to KRA eTIMS. All receipts prominently state: *'Internal receipt — not a KRA eTIMS tax invoice'*.

2. SYSTEM ARCHITECTURE & PLATFORM TOPOLOGY

Cellar POS employs a reactive, single-page application (SPA) architecture combined with platform-native loopback wrappers for Windows Desktop, Android mobile devices, and Apple iOS.

Architecture Layer	Primary Technology	Production Functionality
Frontend Core	React 18 + TypeScript	Strict type safety, reactive state tree, fast virtual DOM rendering
Styling & Thermal	Tailwind CSS + @media print	Dark UI theme and zero-margin receipt formatting for 58mm/80mm printers
Visual Analytics	Recharts (SVG Engine)	Interactive weekly revenue charts categorized by Cash vs M-Pesa
Storage Engine	SQLite-Compatible Local Store	Persistent relational entities, JSON backup/restore with schema validation
Offline Cache	Service Worker + Workbox PWA	Precached static bundles, offline manifest, instant standalone launch

Windows App	PowerShell Local WebServer	Standalone borderless app launcher without requiring Node or Python
Android App	Android Studio + WebView / WebAPK	Native Android wrapper package + 1-tap browser WebAPK installation
Apple iOS App	.mobileconfig + Xcode Swift	Managed WebClip configuration profile and native Swift WKWebView project

3. DATA ARCHITECTURE & RELATIONAL SCHEMA

All application state is maintained in relational entities with strict validation:

- **Product Entity (Product):** Unique SKU records where bottle volumes (250ml, 750ml) have independent pricing and inventory counts.
- **Sale Entity (Sale):** Immutable transaction records with sequential receipt numbering, timestamp, line items, and payment details.
- **Line Item Entity (SaleItem):** Historical snapshot of sold item names, categories, and unit prices at transaction time.
- **Stock Adjustment Entity (StockAdjustment):** Audited inventory adjustment events categorized into Restock, Breakage, Spillage, Correction, and Other.
- **Settings Entity (Settings):** Store profile, business till numbers, threshold configurations, and category taxonomy.

4. FUNCTIONAL SUBSYSTEMS & OPERATIONAL FLOWS

4.1 Till Checkout Flow: Cashiers perform sales through a high-contrast touchscreen interface. The cart prevents negative inventory by locking increments at the current stock level. During checkout, tender selection provides one-tap cash denomination helpers (+100, +200, +500, +1000, +2000, Exact Total) with real-time change calculation, or an M-Pesa transaction reference field for mobile money settlement.

4.2 Reversible Void & Refund Architecture: Transactions are never erased. Voiding a sale creates an audited reversing entry, restores the exact line item quantities back into the physical inventory balance on hand, and stamps the receipt in red: '*** VOIDED / REFUND RECEIPT ***' with mandatory reason tracking.

4.3 Daily Till Reconciliation & Weekly Revenue Analytics: Shift close-out displays the total net revenue, physical Cash in Till Drawer count (amount and transaction count), and Safaricom M-Pesa Statement count. The Recharts weekly revenue graph renders daily cash and digital revenue performance, allowing managers to tap any day's bar to immediately inspect transactions.

5. INSTALLATION & DEPLOYMENT OPTIONS

- **Windows 10/11 Desktop:** Download cellar-pos-desktop-offline-installer.zip (320 KB). Extract and double-click start-cellar-pos.bat to launch in borderless window mode.
- **Android Phone & Tablet:** Open till URL in Chrome/Edge on phone and tap 'Install App', or download the Android Studio native project cellar-pos-android-project.zip.
- **Apple iOS (iPhone/iPad):** Download cellar-pos.mobileconfig in Safari on iPhone to install the Apple WebClip profile, or download the Xcode project cellar-pos-ios-project.zip.

6. DISASTER RECOVERY & HARDWARE INTEGRATION

Daily JSON backups can be downloaded in one click to a USB flash drive from Settings. Compatible with standard USB/Bluetooth 58mm/80mm ESC/POS thermal receipt printers, USB HID barcode scanners, and pulse cash drawers.